

Unit

1

Lesson

3

Spark The Challenge!

Exploring Smart Glasses.

Time Frame:[90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Use the Engineering Design Process (EDP) to brainstorm problems related to vision loss and generate ideas to solve them using assistive technology.
- Identify the role of a Vision Scientist and explain how their work helps people understand, protect, and improve eyesight.
- Differentiate between normal glasses and smart glasses.

Challenge questions of the lesson.

1. What daily challenges do people with vision loss face that affect how they live or move around?
2. How does a Vision Scientist help us understand and protect our eyes?
3. What makes smart glasses different from regular glasses, and how do they help people?
4. If you could invent your own pair of smart glasses, what special feature would you add to support someone?

SEL Relation

- **Community Awareness:** Understand how smart technology, like our Smart Bin, fits into community waste management for a healthier environment.
- **Responsibility:** Recognize our personal role in properly sorting trash, recycling, and contributing to a cleaner school and planet.
- **Collaboration:** Work together during the coding game, discussions, and poster creation to foster a cleaner, more sustainable environment.
- **Appreciation:** Value smart technologies and the resources saved through effective recycling and repurposing.
- **Self-Awareness:** Reflect on learning about smart waste management and identify areas for personal growth in environmental actions.
- **Communication:** Clearly express ideas about smart waste solutions through speaking, drawing, and writing.
- **Critical Thinking & Problem-Solving:** Analyze waste problems, develop solutions through the coding game, and creatively address waste management.
- **Environmental Literacy/Responsibility:** Deepen understanding of waste's environmental impact and take action through smart waste management and recycling.

Engage



Teacher Role:

- **Spark Curiosity:** Introduce the scene with the dialogue by reading it aloud or having students act it out. Highlight the importance of imagination and asking questions in invention.
- **Prompt Career Exploration:** Present the “Career Readiness” scenario. Ask open-ended questions like “What kind of expert could help in this situation?” before showing the video.
- **Model Inquiry Thinking:** Help students reflect on problems related to poor eyesight and how professionals solve them.
- **Bridge to the Big Idea:** Connect the character dialogue and video to the real-world challenge—how we help people with vision problems using smart solutions.

Expected from Students:

- Listen to or read the classroom dialogue and identify the key ideas suggested by the characters about smart glasses and their possible functions.
- Participate in a short class discussion about how inventions often begin with asking questions and using imagination.
- Reflect on the blurry vision scenario and write down who might help solve that problem (e.g., a vision scientist or eye doctor).
- Watch the linked video to discover more about vision-related careers and connect them to the upcoming lesson challenge.
- Begin to consider problems people with vision loss might face and how technology could help, setting the stage for deeper exploration.

Parts of the books included.

Pyramakerz®

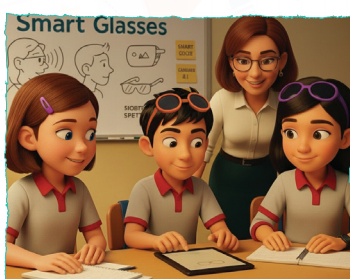
Later that day, in the classroom, Adam and Laila and Jamila sat around a table, sketching and writing down ideas with the help of Mrs. Sara.

Adam tapped his pencil. “What about a device that detects obstacles?”

Laila nodded. “Or something that gives audio directions?”

She grabbed a marker and wrote on the board. “Let’s list everything the glasses should do.”

Ms. Sara smiled. “Great thinking! That’s how real inventors start—with questions and imagination.”



Career Readiness



Imagine you’re wearing a pair of glasses that don’t help you see better. In fact, everything still looks blurry!

Who might be able to figure out what’s going on?

.....

.....

.....

Discover more about Optical Technology through the link.
<https://www.youtube.com/watch?v=hM-1QRvapCs>



Figure 10: Optical Technology.



Explains

Story



1. Start with a Warm Welcome and Story Recap:

- “Good morning, inventors! Last time, we started talking about how Jamila wanted to help her blind grandfather. Do you remember who else was with her?”
- (Encourage a few students to recall the characters: Adam, Laila, Jamila—and how they were curious about vision problems.)
- “Do you remember what problem they were trying to solve? That’s right—how to help someone who can’t see well move around safely!”

2. Continue the Story with Expression:

- “Today, we’re back in the classroom with our team of young inventors. They’re full of ideas... and guess what? Mrs. Sara is ready to help them turn their imagination into something real!”
- (Read the story dialogue with expression—use tone for Adam’s curiosity, Laila’s excitement, and Mrs. Sara’s encouragement.)
- “Let’s listen closely to their conversation and see what they come up with!”

Possible Strategies for Implementation:

- **Think-Pair-Share:** After the story, ask students to individually think about features smart glasses could have, pair up with a classmate to exchange ideas, then share with the class.
- **Role Play:** Invite a few students to act out the story with simple props, encouraging empathy and imagination. This helps students connect emotionally with the problem.
- **Idea Web Mapping:** On the board or paper, create an idea web titled “What Should Smart Glasses Do?” Students contribute features like obstacle detection, audio directions, or image zooming.
- **Inventor’s Notebook:** Give students a simple “Inventor’s Log” page to sketch their own version of smart glasses and write what problems they want to solve.
- **Use SEL Prompts:** Ask, “How do you think a blind person feels walking alone?” or “How would you feel if you couldn’t see well?” to activate empathy and deepen purpose.

Career Readiness



Objective:

Students will explore the real-world role of a Vision Scientist by imagining a scenario of visual impairment, predicting who could help diagnose the issue, and connecting the scenario to an actual career in eye health through a short video. This helps them understand how science connects to real careers that support well-being and vision care.

Possible Student Answer:

- "An eye doctor or a vision scientist might be able to figure out what's going on because they know how to test eyes and understand why glasses may not work."

Possible Strategies for Implementation:

- **Scenario Walkthrough:** Begin by reading the prompt aloud and guiding students through the imaginative scenario. Ask: "Have you ever worn glasses that didn't help? How did that feel?"
- **Think-Pair-Share:** Let students brainstorm individually, then pair up to share their thoughts on who could help and why.
- **Video Integration:**
- **Activity Link:** <https://www.youtube.com/watch?v=hM-1ORvapCs>
 - Before playing, ask them to listen for what the Vision Scientist does and why their work is important.
- **Discussion Prompt:** After the video, facilitate a short discussion: "What tools or knowledge does the Vision Scientist use? How do they help people?"
- **Career Chart Connection:** Add the Vision Scientist to a class "STEM Careers" wall or poster, noting their tools, workplace, and how they help others.



Explain



Teacher Role:

- Before the Video:
 - Activate prior knowledge by revisiting the story where students brainstormed smart glasses ideas.
 - Set the purpose for viewing: "Let's find out how real engineers use a special process—called the EDP—to solve problems like vision loss."
- During the Video:
 - Pause the video at key points to check for understanding (e.g., Define → Imagine → Plan...).
 - Ask prompting questions: "What's happening here? Why do you think that step is important?"
- After the Video:
 - Facilitate a short discussion: "Which step of the EDP do you think is the most important when designing for someone who's blind?"
 - Record students' reflections on the board or on sticky notes to display the EDP visually.

Expected From Students:

- Watch and understand the EDP steps shown in the video.
- Identify and explain each step using their own words.

- Connect the EDP process to their previous ideas about smart glasses or assistive vision tools.
- Begin preparing to apply EDP in their own prototype later in the unit.

Parts of the books included.

Screen Lessons

"Now it's time to be a helper and use our problem-solving steps (EDP) to help people with vision disorders and blindness.

https://www.youtube.com/watch?v=MAhpfFt_mWM&t=1s



Screen Lessons

Objective:

Students will learn how to apply the Engineering Design Process (EDP) to develop assistive technologies for people with vision impairments. Through a guided video, they will explore real-world challenges and how innovative thinking can create inclusive solutions.

Video link: https://youtu.be/MAhpfFt_mWM?si=n22-IDi7NAbFMTpG



Possible Strategies for implementation:

1. Pre-Video:

- Use a KWL chart (What I Know – What I Want to Know – What I Learned) about EDP.
- Ask: "Have you ever had a problem that you solved step by step? Let's compare that to how engineers solve problems."

2. During Video:

- Pause after each EDP step and write the name of the step on the board.
- Ask students to sketch a symbol for each step (e.g., lightbulb for "Imagine," checklist for "Plan").

3. Post-Video:

- Group students to discuss how they might use EDP to help someone with a vision disorder.
- Create an EDP anchor chart together using ideas from the video and previous discussions.

Engineering Design Process

Name:

Project:

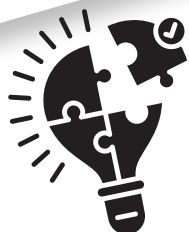
Ask

- What problems do people with vision loss face in their daily lives?
- Why don't regular glasses help everyone see better?
- What features might someone who is blind or visually impaired need in assistive glasses?
- What are some technologies we already know that can help people see, hear, or move around?



Imagine

- What if we made glasses that give voice directions to help someone find their way?
- Could we use sensors to detect obstacles or danger?
- What if the glasses could connect to a phone or app to help the person navigate or call for help?
- Let's brainstorm different tools we could combine—like buzzers, lights, or GPS—and sketch out what the glasses might look like.



Plan

- Choose one main problem your glasses will solve (e.g., obstacle detection, voice directions, or alerts).
- Decide what parts or features you want to include in your smart glasses (sensor, speaker, button, LED, etc.).
- Sketch the design of your smart glasses and label the parts.
- Write the steps you'll follow to start building a prototype.



Create



Improve



Explorer's Toolkit



Teacher Role:

- Introduce the activity by explaining the difference between dark glasses and smart glasses and how they each serve different needs.
- Guide students through reading and categorizing the features in the prompt list.
- Encourage critical thinking by asking guiding questions like:
 - “Which features sound like they require technology?” or
 - “Which ones are more about basic protection?”
- Support struggling students by reading features aloud or helping with decoding technical words.
- Circulate and provide support, clarification, or rephrasing of feature statements as needed.
- Facilitate discussion at the end to compare categorizations and clarify misconceptions.

Expected from Students:

- Read each eyewear feature statement and determine whether it best fits dark glasses or smart glasses.
- Record their sorting choices in the correct column on the worksheet.
- Justify their thinking through group discussion and reflection.
- Identify key differences between assistive and protective eyewear.

Parts of the book included:



DIY Lab

Objective:

Students will analyze and classify features of two types of eyewear—dark glasses and smart glasses—to understand the distinction between basic protection tools and advanced assistive technology. Through this comparison, they will explore how technology can enhance independence and accessibility for people with visual impairments.

Activity Instructions:

1. Read the list of features provided at the top of the page.
2. With a partner or small group, discuss each feature and decide whether it applies to:
 - Dark Glasses (protection, non-tech).
 - Smart Glasses (assistive technology, navigation).
3. Write each feature under the correct category in the provided columns.
4. Review your choices and explain your sorting to the class.

Possible Strategies for Implementation:

1. Pre-Activity Strategy:

- Use a “See–Think–Wonder” visible thinking routine by showing an image of both glasses types and asking:
 - “What do you see? What do you think these do? What do you wonder about them?”

2. During Activity Strategy:

- Implement Kagan’s RallyCoach – In pairs, students take turns reading each feature aloud and coaching each other to decide where to categorize it. They must agree before writing it down.

3. Post-Activity Strategy:

- Use Think–Pair–Share to allow students to reflect:
 - “Which feature was hardest to decide on? Why?”
- Then share with the class and check for misconceptions.

4. UDL Support:

- Provide printed symbols (e.g., eye icon, tech icon) beside each feature for ELL or struggling readers. Allow some students to sort using cards physically before writing.



Now I Can

Objective:

Students will reflect on their learning journey by identifying and expressing what they now understand about assistive technology for vision support, the role of vision scientists, and the differences between regular and smart glasses. Through self-assessment using a “Now I Can” checklist, students will demonstrate their understanding and confidence in applying the Engineering Design Process to real-world problems.

Activity Instructions:

1. Distribute the “Now I Can” Reflection Page to each student, or if they have the book, let them open the page.
2. Ask students to read each “Now I Can” statement aloud silently or with a partner.
3. Students should tick each box where they feel confident in their understanding and add a quick example or explanation underneath, if time permits.

4. Allow 5–3 minutes for quiet individual reflection.
5. Then facilitate a group or partner discussion where students share one thing they're proud of understanding.
6. Optionally, collect the pages to formatively assess comprehension and address any unclear areas in future lessons.

Relation to Standards.

1. NGSS (Next Generation Science Standards).

- **3-5-ETS1-1:** Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.
 - Students identify vision-related challenges and apply the Engineering Design Process to generate assistive tech solutions.

2. ISTE Standards for Students.

- **1.4 Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions.
 - Learners design solutions (like smart glasses) to address vision difficulties through creative thinking and planning.
- **1.5 Computational Thinker:** Students develop and employ strategies for understanding and solving problems.
 - Students analyze problems like navigation or access for the blind, break them down, and develop feasible solutions.

3. Common Core ELA Standards.

- **CCSS.ELA-LITERACY.SL.3.1:** Engage effectively in a range of collaborative discussions.
 - Students brainstorm, share ideas, and explain their thinking with peers when listing and evaluating eyewear features.
- **CCSS.ELA-LITERACY.RI.3.3:** Describe the relationship between a series of events, ideas, or concepts.
 - Students explain how vision scientists help society and how smart vs. regular glasses serve different functions.

4. CSTA Computer Science Standards

- **1B-AP-12:** Modify, remix, or reuse parts of programs to create a new program.
 - If extended into coding (e.g., coding smart glasses features), students apply this through reuse of sensor-based logic.